DC Circuits — Reference

Textbook: Ulaby, Maharbiz, Furse, *Circuit Analysis and Design*, 3rd ed., chapters 1–2. Gray tags like [book 2-1] point to the book section. The book writes voltage as υ (upsilon). These notes write v.

prefixes

Symbol	Prefix	Power	Multiply by	In words
E	exa	10^{18}		quintillion
P	peta	10^{15}		quadrillion
T	tera	10^{12}	1,000,000,000,000	trillion
G	giga	10^{9}	1,000,000,000	billion
M	mega	10^{6}	1,000,000	million
k	kilo	10^{3}	1,000	thousand
—	(none)	10^{0}	1	one
m	milli	10^{-3}	0.001	thousandth
μ	micro	10^{-6}	0.000 001	millionth
n	nano	10^{-9}	0.000 000 001	billionth
p	pico	10^{-12}	0.000 000 000 001	trillionth
f	femto	10^{-15}		quadrillionth
a	atto	10^{-18}		quintillionth

Capital M (mega) and small m (milli) are 10^{9} apart. [book Table 1-2]

quantities and units

Quantity	Symbol	Unit	In base units
charge	q, Q	coulomb (C)	A·s
current	i, I	ampere (A)	C/s
voltage	v, V (book: υ)	volt (V)	J/C
power	p, P	watt (W)	J/s = V·A
energy	w, W	joule (J)	W·s
resistance	R	ohm (Ω)	V/A
conductance	G	siemens (S)	A/V $= 1/\Omega$
resistivity	ρ (rho)	ohm-meter	Ω·m
conductivity	σ (sigma)	siemens/meter	S/m
capacitance	C	farad (F)	C/V
inductance	L	henry (H)	V·s/A
frequency	f	hertz (Hz)	1/s
length, area	ℓ, A	m, m^2	—
time	t	second (s)	—
temperature	T	kelvin (K), °C	K =°C+273

Lowercase (q, i, v, p, w) can change with time. Uppercase (Q, I, V, P, W) is a constant (dc). [book Table 1-1]

constants and conversions

Item	Value
charge of one electron	$-e$, $e = 1.6 \times 10^{-19}$ C (more exact: 1.602)
charge of one proton	$+e$
electrons in 1 C	6.25×10^{18}
minute, hour	1 min = 60 s, 1 h = 3600 s
ampere-hour	1 Ah = 3600 C, 1 mAh = 3.6 C
watt-hour, kilowatt-hour	1 Wh = 3600 J, 1 kWh $= 3.6 \times 10^{6}$ J
length	1 mm $= 10^{-3}$ m, 1 cm $= 10^{-2}$ m
area	1 $mm^2 = 10^{-6}$ m^2, 1 $cm^2 = 10^{-4}$ m^2
specific heat of water	$c = 4186$ J/(kg·°C)
mass of water	1 L ≈ 1 kg, 1 fl oz ≈ 0.02957 kg

The book uses $e = 1.6 \times 10^{-19}$ C. Use it to match the book's answers.

circuit terminology

Term	Meaning [book Table 1-4]
node	all the wire between elements: one voltage. Points joined by a wire or a short are the same node
ordinary node	connects to 2 elements
extraordinary node	connects to 3 or more elements
branch	one element between two nodes
path	a chain of branches that repeats no node
extraordinary path	a path between two adjacent extraordinary nodes
loop	a closed path
independent loop	a loop with at least one branch that no other chosen loop has
mesh	a loop that encloses no other loop
in series	same current: only ordinary nodes between them
in parallel	same voltage: they share both end nodes
planar circuit	can be drawn with no crossing wires. A crossing without a dot is not a connection
open circuit	$i = 0$, $R = \infty$, any voltage
short circuit	$v = 0$, $R = 0$, any current
ground	the 0 V reference node. Chassis ground: the metal frame. Earth ground: a rod in the soil

conductivity and resistivity, 20°C

Material	σ (S/m)	$\rho = 1/\sigma$ (Ω·m)
Conductors		
silver	6.17×10^{7}	1.62×10^{-8}
copper	5.81×10^{7}	1.72×10^{-8}
gold	4.10×10^{7}	2.44×10^{-8}
aluminum	3.82×10^{7}	2.62×10^{-8}
iron	1.03×10^{7}	9.71×10^{-8}
mercury (liquid)	1.04×10^{6}	9.58×10^{-7}
Semiconductors		
carbon (graphite)	7.14×10^{4}	1.40×10^{-5}
pure germanium	2.13	0.47
pure silicon	4.35×10^{-4}	2.30×10^{3}
Insulators		
paper	$\sim 10^{-10}$	$\sim 10^{10}$
glass	$\sim 10^{-12}$	$\sim 10^{12}$
teflon	$\sim 3.3 \times 10^{-13}$	$\sim 3 \times 10^{12}$
porcelain	$\sim 10^{-14}$	$\sim 10^{14}$
mica	$\sim 10^{-15}$	$\sim 10^{15}$
polystyrene	$\sim 10^{-16}$	$\sim 10^{16}$
fused quartz	$\sim 10^{-17}$	$\sim 10^{17}$
Common materials		
distilled water	5.5×10^{-6}	1.8×10^{5}
drinking water	$\sim 5 \times 10^{-3}$	~ 200
sea water	4.8	0.2
rubber	10^{-13}	10^{13}
Biological tissues		
blood, muscle	~ 1.5	~ 0.67
fat	~ 0.1	10

Small ρ (big σ) = good conductor. [book Table 2-1] If a problem gives ρ or σ, use the problem's value.

wire size (AWG) and cross-section area

AWG	Diameter d (mm)	Area $A = \pi d^2/4$ (m^2)
0	8.3	5.41×10^{-5}
2	6.5	3.32×10^{-5}
4	5.2	2.12×10^{-5}
6	4.1	1.32×10^{-5}
10	2.6	5.31×10^{-6}
14	1.6	2.01×10^{-6}
18	1.0	7.85×10^{-7}
20	0.8	5.03×10^{-7}

Bigger AWG number = thinner wire. [book Table 2-2]

Formula	In words
$A = \pi r^2 = \dfrac{\pi d^2}{4}$	round wire: (area) $= \pi$(radius)^2
$d = \sqrt{\dfrac{4A}{\pi}}$	(diameter) from the area
$A = w \cdot h$	rectangle: (width)(thickness), the face the current passes *through*
$A = \pi(b^2 - a^2)$	hollow tube: outer radius b, inner radius a

Convert mm to m before you square: $(1\text{ mm})^2 = 10^{-6}\text{ m}^2$. ℓ is along the current, A is across it. For a block, the current direction decides which side is ℓ.

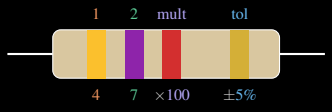
resistor color code

	Color	Digit	Multiplier	Tolerance	Temp. coef.
	black	0	$\times 1$	—	—
	brown	1	$\times 10$	$\pm 1\%$	100 ppm
	red	2	$\times 100$	$\pm 2\%$	50 ppm
	orange	3	$\times 1$k	—	15 ppm
	yellow	4	$\times 10$k	—	25 ppm
	green	5	$\times 100$k	$\pm 0.5\%$	—
	blue	6	$\times 1$M	$\pm 0.25\%$	10 ppm
	purple (violet)	7	$\times 10$M	$\pm 0.1\%$	5 ppm
	gray	8	—	—	—
	white	9	—	—	—
	gold	—	$\times 0.1$	$\pm 5\%$	—
	silver	—	$\times 0.01$	$\pm 10\%$	—

[book Fig. 2-2] Hold it so the wider gap is before the last band(s), on the right. Read left to right.

Formula	In words
4 bands: $R = (b_1 b_2) \times 10^{b_4} \pm b_5$	(two digits) $\times$ (multiplier) $\pm$ (tolerance)
5 bands: $R = (b_1 b_2 b_3) \times 10^{b_4} \pm b_5$	(three digits) $\times$ (multiplier) $\pm$ (tolerance)
6 bands: add b_6	5-band reading plus the temperature coefficient, ppm/°C

The multiplier band is the number of zeros. Digits 2–7 follow the rainbow: red, orange, yellow, green, blue, purple.



Worked: read the resistor above

1. PIECES (WIDE GAP ON THE RIGHT, READ LEFT TO RIGHT)
yellow $\to 4$, purple $\to 7$, red $\to \times 100$, gold $\to \pm 5\%$

↓

2. TEMPLATE (4 BANDS)
$R = (\boxed{\text{digit 1}}\boxed{\text{digit 2}}) \times \boxed{\text{multiplier}} \pm \boxed{\text{tolerance}}$

↓

3. PLACE
$R = (47) \times 100 \pm 5\% = 4.7\text{ k}\Omega$

↓

4. TOLERANCE RANGE
$4700 \times 0.05 = 235$: from 4465 to 4935 Ω

↓

5. CHECK (REVERSE)
$4700 = 47 \times 10^2$: digits 4, 7, then 2 zeros: yellow, purple, red ✓

Quick example	Work $\to$ answer
brown, black, orange, gold	$10 \times 10^3 = 10\text{ k}\Omega$, $\pm 5\%$
5 bands: brown, black, black, red, brown	$100 \times 10^2 = 10\text{ k}\Omega$, $\pm 1\%$
red, red, gold, gold	$22 \times 0.1 = 2.2\,\Omega$, $\pm 5\%$
(reverse) $330\,\Omega$, $\pm 5\%$	33×10^1: orange, orange, brown, gold

formula sheet (symbols only)

2. charge, current $Q = It$, $i = \dfrac{dq}{dt}$, $\Delta Q = \int_{t_1}^{t_2} i\,dt$, $n = \dfrac{|Q|}{e}$

3. voltage $v_{ab} = \dfrac{dw}{dq}$, $V_{ab} = V_a - V_b$, $W = QV$

4. power, energy $p = vi$, $\sum p = 0$, $w = \int p\,dt$, $W = Pt$, $P_{av} = \dfrac{1}{T}\int_0^T p\,dt$

5. extras $\text{Wh} = V \cdot \text{Ah}$, $t = \dfrac{\text{Ah}}{I}$, $E = cm\Delta T$, $\text{kWh} = \dfrac{\text{J}}{3.6 \times 10^6}$

6. sources $V_L = \dfrac{R_L}{R_s + R_L} v_s$. dependent: αv_x, $r i_x$, $g v_x$, βi_x

7. resistors $V = IR$, $R = \dfrac{\rho \ell}{A} = \dfrac{\ell}{\sigma A}$, $P = I^2 R = \dfrac{V^2}{R}$, $R = R_0(1 + \alpha T)$

8. KCL, KVL $\sum i_{\text{in}} = \sum i_{\text{out}}$, $\sum v = 0$. N unknowns: M KVL + $(N - M)$ KCL

9. combine $R_1 + R_2$, $\dfrac{R_1 R_2}{R_1 + R_2}$, $G_{eq} = \sum G$

10. dividers $v_1 = \dfrac{R_1}{R_1 + R_2} v_s$, $i_1 = \dfrac{R_2}{R_1 + R_2} i_s = \dfrac{G_1}{G_{eq}} i_s$

11. source transf. $i_s = \dfrac{v_s}{R_s}$, $v_s = i_s R_s$ (same R_s)

12. $\Delta \to Y$ $R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$, $R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$

13. bridge $R_x = \dfrac{R_2}{R_1} R_3$, $V_{out} \approx \dfrac{V_0}{4} \dfrac{\Delta R}{R}$, $R = R_0(1 + \alpha P)$

14. diode, LED forward: $I = \dfrac{V_s - V_F}{R}$, $V_F \approx 0.7$ V (Si)

calculus and algebra tools

Derivatives (a, ω, n, c constants):

Function	Derivative
c, t^n	0, nt^{n-1}
e^{at}	ae^{at}
$\sin(\omega t)$, $\cos(\omega t)$	$\omega\cos(\omega t)$, $-\omega\sin(\omega t)$
$f\cdot g$ (product)	$f'g+fg'$
$f(g(t))$ (chain)	$f'(g)\cdot g'$. For example, $\sin^2(\omega t)\to 2\omega\sin(\omega t)\cos(\omega t)$

Integrals:

Function	Antiderivative
t^n	$\dfrac{t^{n+1}}{n+1}$
e^{at}	$\dfrac{e^{at}}{a}$
$\sin(\omega t)$, $\cos(\omega t)$	$-\dfrac{\cos(\omega t)}{\omega}$, $\dfrac{\sin(\omega t)}{\omega}$
$e^{at}\cos(bt)$	$\dfrac{e^{at}\left(a\cos bt+b\sin bt\right)}{a^2+b^2}$
$e^{at}\sin(bt)$	$\dfrac{e^{at}\left(a\sin bt-b\cos bt\right)}{a^2+b^2}$
$\int_0^\infty Ae^{-at}\,dt$	$\dfrac{A}{a}$ $(a>0)$

Definite integral: $F(\text{top})-F(\text{bottom})$. $e^0=1$. $e^{-at}\to 0$ as $t\to\infty$.
Trig identities: $\sin x\cos x=\frac{1}{2}\sin 2x$, $\cos^2 x=\frac{1}{2}(1+\cos 2x)$, $\sin^2 x=\frac{1}{2}(1-\cos 2x)$.
Maximum or minimum: set the derivative to 0 and solve for t. Example: largest q where $\frac{dq}{dt}=0$, and the largest p where $\frac{dp}{dt}=0$.

Solving simultaneous equations

2 unknowns ($a_1x+b_1y=c_1$, $a_2x+b_2y=c_2$), Cramer's rule:

$$D=a_1b_2-a_2b_1 \qquad x=\frac{c_1b_2-c_2b_1}{D} \qquad y=\frac{a_1c_2-a_2c_1}{D}$$

3 unknowns:

$$D=\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=a_1(b_2c_3-b_3c_2)-b_1(a_2c_3-a_3c_2)+c_1(a_2b_3-a_3b_2).$$

$x_k=\dfrac{D_k}{D}$, where D_k is D with column k replaced by the right-hand sides. Or: eliminate one unknown at a time, or use a calculator's matrix solver (rref). [book App. B]

Quick example	Work → answer
$3x+2y=12$, $x-y=-1$	$D=3(-1)-1(2)=-5$. $x=\frac{12(-1)-(-1)(2)}{-5}=2$, $y=\frac{3(-1)-1(12)}{-5}=3$

which tool? — given → find

The number in parentheses is the concept section.

Given → find	Do this	
a circuit → nodes, series, parallel	terminology table	(1)
charge Q → electrons	$n=\lvert Q\rvert/e$	(2)
$q(t)\to i$ / $i(t)\to\Delta Q$	differentiate / integrate	(2)
a $q(t)$ or $i(t)$ graph	slope / area	(2)
node voltages, new ground	subtract the new ground's old value	(3)
V, I → power, sign	$P=VI$, passive sign convention	(4)
$v(t)$, $i(t)$ → energy, average power	$\int vi\,dt$ / $\frac{1}{T}\int p\,dt$	(4)
battery, kWh, heat, cost	extras	(5)
switch at t_0	redraw: closed = short, open = open	(6)
two of V, I, R → the third	Ohm's law	(7)
wire: $R\leftrightarrow\ell$, A, material	$R=\rho\ell/A$ and the tables	(7)
resistance at a new temperature	$R=R_0(1+\alpha T)$	(7)
color bands → R	color chart (ref)	
currents at a node / voltages in a loop	KCL / KVL	(8)
N unknown currents	M KVL + $(N-M)$ KCL, solve	(8)
network → R_{eq}	combination chart	(9)
series → v share / parallel → i share	dividers	(10)
mixed sources and resistors	source transformation	(11)
not series, not parallel (a bridge)	$\Delta\leftrightarrow Y$	(12)
bridge balance / sensor output	Wheatstone	(13)
force on a sensor → V_{out}	piezoresistor, cantilever	(13)
diode or LED in a circuit	assume on, check $I>0$	(14)

DC Circuits — Concepts

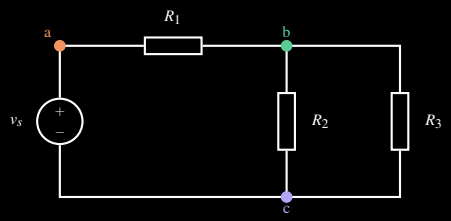
How to read a worked example. Every example is a puzzle: a colored piece in the problem or the diagram goes in the slot of the same color.

Step	What it does
1. PIECES	Read each value off the problem or diagram. Give each one a color. Where to stand or what to read first is named in this title.
2. TEMPLATE	The formula, with an empty slot [like this] for each piece, in the piece's color.
3. PLACE	Put each piece in its slot. Then do the arithmetic.
middle steps	One operation each, named by the operation: SOLVE, DROPS, REVERSE.
last: CHECK	Test the answer: KCL, KVL, power balance, or a reverse calculation.

Colors. Step 1 sets the colors for each example. In a diagram, a colored element goes in the slot of its color.

1. reading a circuit [book 1-3]

Label the nodes first. Every wire segment between elements is one node, even if it is drawn long. Merge points joined by a wire or a short.



Worked: classify the circuit above

1. PIECES (WHAT TOUCHES EACH NODE)
a: v_s, R_1 (2) b: R_1, R_2, R_3 (3) c: v_s, R_2, R_3 (3)

↓

2. TEMPLATES
series: only an [ordinary node (2 elements)] between them
parallel: they share [one end node] and [the other]

↓

3. PLACE
a ordinary ⇒ v_s, R_1 series. R_2, R_3 share b and c ⇒ parallel.
Meshes: (v_s, R_1, R_2) and (R_2, R_3) = 2.

↓

4. CHECK
meshes = branches − nodes $+1 = 4-3+1 = 2$ ✓

Meters [book Fig. 1-18]: a voltmeter goes in parallel (across the element), red lead on +. Ideal: no current. An ammeter goes in series (in the path). Ideal: no drop.
Same branch, same direction = same current. Opposite arrows in one branch are negatives.

2. charge and current [book 1-4]

Current is the flow of positive charge. Electrons ($-e$) moving one way count as current the other way. Protons ($+e$) count in their own direction.

Formula	In words
$n = \frac{\lvert Q\rvert}{e}$, $Q = ne$	(electrons) = (charge) ÷ (charge of one)
$Q = It$, $I = \frac{Q}{t}$, $t = \frac{Q}{I}$	(charge) = (current)(time), constant current
$i(t) = \frac{dq}{dt}$	(current) = rate of change of (charge): differentiate
$q(t) = \int_{-\infty}^{t} i\,dt$	(charge so far) = total of (current): integrate
$\Delta Q = q(t_2) - q(t_1) = \int_{t_1}^{t_2} i\,dt$	(charge moved from t_1 to t_2)
graph of $q(t)$	current = slope of each straight piece
graph of $i(t)$	charge = area under it (piece by piece)

Quick example	Work → answer
2 mC → electrons	$\frac{2\times10^{-3}}{1.6\times10^{-19}} = 1.25\times10^{16}$
3×10^{18} electrons/s go $+x$ / 10^{18} protons/s go $-x$	both are current toward $-x$: $4\times10^{18}\cdot1.6\times10^{-19} = 0.64$ A in $-x$
90 C in 30 s / / 0.5 A for 2 min	3 A / / $0.5\cdot120 = 60$ C
$q = 3t^2$ mC → i	$6t$ mA
$q = 0.8(1-e^{-0.5t})$ nC → i	$0.8\cdot0.5e^{-0.5t} = 0.4e^{-0.5t}$ nA
q rises 0 → 6 C in a line over 0–3 s	slope: $6/3 = 2$ A
$i = 4$ A for 2 s, then 0	area: $4\cdot2 = 8$ C
$q = 10te^{-t}$ C, when is q largest?	$\frac{dq}{dt} = 10e^{-t}(1-t) = 0 \Rightarrow t = 1$ s, $q = 10/e = 3.68$ C
$i = 0.4e^{-0.5t}$ nA, total to ∞	$\frac{A}{a} = \frac{0.4}{0.5} = 0.8$ nC

Time in seconds. Given charge, find current: differentiate. If your answer has a higher power of t, you integrated.

Worked: (integrate) $i(t) = 20e^{-0.5t}$ mA from $t = 0$ to 4 s

1. PIECES
$A = 20$ mA $a = -0.5$ limits 0, 4

↓

2. TEMPLATE
$\Delta Q = \int_{\boxed{\text{bottom}}}^{\boxed{\text{top}}} \boxed{A}e^{\boxed{a}t}\,dt = F(\text{top}) - F(\text{bottom}), \quad F = \frac{\boxed{A}}{\boxed{a}}e^{\boxed{a}t}$

↓

3. PLACE
$F(t) = \frac{20}{-0.5}e^{-0.5t} = -40e^{-0.5t}$

↓

4. TOP MINUS BOTTOM
$F(4) - F(0) = -40e^{-2} - (-40) = -5.413 + 40 = 34.59$ mC *($mA\cdot s = mC$, $e^0 = 1$)*

↓

5. CHECK
all the way to ∞ gives $\frac{A}{\lvert a\rvert} = 40$ mC. $34.59 < 40$, most of it by 4 s ✓

Worked: (product rule) $q(t) = 0.5t\cos(100\pi t)$ μC

1. PIECES
$f = 0.5t$, $f' = 0.5$ $g = \cos(100\pi t)$, $g' = -100\pi\sin(100\pi t)$

↓

2. TEMPLATE
$i = \frac{dq}{dt} = \boxed{f'}\boxed{g} + \boxed{f}\boxed{g'}$

↓

3. PLACE
$i = 0.5\cos(100\pi t) + 0.5t\left(-100\pi\sin(100\pi t)\right) = 0.5\cos(100\pi t) - 50\pi t\sin(100\pi t)$ μA

↓

4. CHECK
$t = 0$: $i = 0.5$ μA. Near 0, $q \approx 0.5t$, slope 0.5 ✓

3. voltage, node voltage, and ground [book 1-5.1]

Voltage is energy per charge between two points. It is *across* an element. A **node voltage** is measured from ground (0 V).

Formula	In words
$v_{ab} = \frac{dw}{dq}$	(voltage of a over b) = (energy) per (charge)
$V = \frac{W}{Q}$, $W = QV$	constant case
$V_{ab} = V_a - V_b$	(voltage between two nodes) = (node voltage a) − (node voltage b)
$V_{ab} = -V_{ba}$	reversing the order flips the sign

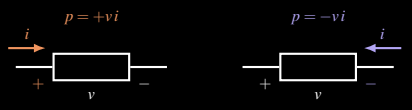
Moving ground shifts every node voltage by the same amount. Voltage *differences* do not change.

Quick example	Work → answer
30 J moves 5 C	$V = 30/5 = 6$ V
$V_a = 9$, $V_b = 5$, $V_c = 0$ (ground c) → ground at b	subtract 5: $V_a = 4$, $V_b = 0$, $V_c = -5$ V
V_{ab} before and after	$9-5=4$ and $4-0=4$: same

4. power and energy [book 1-5.2]

Formula	In words
$p = v\,i$	(power) = (voltage)(current)
$I = \dfrac{P}{V}$, $V = \dfrac{P}{I}$	rearranged
$\sum_k p_k = 0$	conservation: (power supplied) = (power received)
$w = \int_{t_1}^{t_2} p\,dt$	(energy) = total of (power) over time
$W = Pt$, $P = \dfrac{W}{t}$, $t = \dfrac{W}{P}$	constant power
$W = V\,I\,t$	(energy) = (voltage)(current)(time)
$P_{av} = \dfrac{1}{T}\int_0^T p\,dt$	(average power) = (energy over T) $\div T$

Passive sign convention: i enters the $+$ side of $v \Rightarrow p = +vi$. Enters the $-$ side $\Rightarrow p = -vi$. Then $p > 0$: the device receives (absorbs) power. $p < 0$: it supplies power.



Worked: passive sign: who receives, who supplies

1. PIECES
the side the arrow enters: $+$ or $-$, and the values v, i

↓

2. TEMPLATE
$p = \boxed{+}/\boxed{-}\,v\,i$ $\quad p > 0$ receives, $\quad p < 0$ supplies

↓

3. PLACE
5 V, 2 A into $+$: $+(5)(2) = +10$ W receives $\quad$ into $-$: $-(5)(2) = -10$ W supplies
-3 V, 4 A into $+$: $+(-3)(4) = -12$ W supplies

↓

4. CHECK
a resistor always receives: its current enters $+$, so $p = +vi > 0$ ✓

Quick example	Work → answer
three devices: $+20$, -50, p_3	$20-50+p_3=0 \Rightarrow p_3 = 30$ W
24 W for 1 h	$24 \cdot 3600 = 86{,}400$ J
$p = 10t$ W over 0–2 s → P_{av}	$\frac{1}{2}\int_0^2 10t\,dt = \frac{1}{2}(20) = 10$ W
$p(t)$ graph	energy = area under p. Build $p = v \cdot i$ piece by piece
$v = 50(1-e^{-0.5t})$ V, $i = 4e^{-0.5t}$ mA at $t = 1$	$19.67 \cdot 2.426 = 47.7$ mW

$\int v\,i\,dt \neq \int v\,dt \cdot \int i\,dt$. Multiply first. Maximum power: set $\frac{dp}{dt} = 0$.

Worked: total energy, $v = 50(1-e^{-0.5t})$ V, $i = 4e^{-0.5t}$ mA, $0 \to \infty$

1. PIECES
$v = 50(1-e^{-0.5t})$ V $\quad i = 4e^{-0.5t}$ mA $\quad$ limits $0, \infty$

↓

2. TEMPLATE (MULTIPLY FIRST)
$w = \int_{\boxed{\text{bottom}}}^{\boxed{\text{top}}} \boxed{v} \cdot \boxed{i}\,dt$

↓

3. PLACE, MULTIPLY
$p = 50(1-e^{-0.5t}) \cdot 4e^{-0.5t} = 200e^{-0.5t} - 200e^{-t}$ mW $(e^{-0.5t}e^{-0.5t} = e^{-t})$

↓

4. INTEGRATE (SHORTCUT $\int_0^\infty Ae^{-at}\,dt = \frac{A}{a}$)
$w = \dfrac{200}{0.5} - \dfrac{200}{1} = 400 - 200 = 200$ mJ $(V \cdot mA \cdot s = mJ)$

↓

5. CHECK (ANTIDERIVATIVE)
$F = -400e^{-0.5t} + 200e^{-t}$: $\quad F(\infty) - F(0) = 0 - (-200) = 200$ ✓

5. batteries, utility energy, heat, cost

Course-assignment topics. Ah is charge. Wh and kWh are energy.

Formula	In words
$\text{Wh} = V \cdot \text{Ah}$	(energy rating) = (voltage)(charge rating)
$t = \dfrac{\text{Ah}}{I} = \dfrac{\text{Wh}}{P}$	(run time, hours)
$\text{J} = \text{Wh} \cdot 3600$, $\quad \text{C} = \text{Ah} \cdot 3600$	hours to seconds
$n = \dfrac{W}{W_{\text{one}}}$	(number of batteries filled), round down
$E = cm\Delta T$	(heat) = (specific heat)(mass)(temperature change)
$\Delta T = \dfrac{E}{cm}$, $\quad m = \dfrac{E}{c\Delta T}$	rearranged
$\text{cost} = \dfrac{\text{J}}{3.6 \times 10^6} \cdot \text{rate}$	joules → kWh → cost

Quick example	Work → answer
2.5 Ah, 500 mA load	$2.5/0.5 = 5$ h
12 V, 60 Wh, 250 mA load	$P = 3$ W. $\quad 60/3 = 20$ h
1.08 MJ fills 3000 mAh, 3.7 V batteries	one: 11.1 Wh = 39,960 J. $\quad 27.03 \to 27$
0.5 kg water, 20 → 80°C	$4186 \cdot 0.5 \cdot 60 = 125{,}580$ J
125,580 J at 15 cents/kWh	$0.03488 \cdot 15 = 0.523$ cents

6. circuit elements, sources, switches [book 1-6]

Active devices need a power source (transistors, ICs). **Passive** devices do not (resistors, capacitors, inductors).

Element	Rule
ideal voltage source	$v = V_s$ for any current. i–v graph: vertical line
ideal current source	$i = I_s$ for any voltage. i–v graph: horizontal line
real voltage source	v_s in series with R_s. Ignore R_s if $R_s \ll R_L$
real current source	i_s in parallel with R_s. Ignore R_s if $R_s \gg R_L$
SPST switch	closed = short, open = open. The label $t = t_0$ is when it changes
SPDT switch	the common terminal moves from contact 1 to contact 2 at t_0

Formula	In words
$V_L = \dfrac{R_L}{R_s + R_L}\,v_s$	(load voltage) of a real source
$v = \alpha\,v_x$	VCVS: α has no units
$v = r\,i_x$	CCVS: r in V/A = Ω
$i = g\,v_x$	VCCS: g in A/V = S
$i = \beta\,i_x$	CCCS: β has no units

Quick example	Work → answer
9 V, $R_s = 1\,\Omega$, $R_L = 8\,\Omega$	$V_L = \frac{8}{9} \cdot 9 = 8$ V
CCVS $10\,i_x$, $i_x = 0.5$ A	5 V
VCCS $0.2\,v_x$, $v_x = 10$ V	2 A
switch closes at $t = 2$ s	two circuits: $t < 2$ (open branch), $t > 2$ (short)

Dependent sources are models, not energy sources. Solve them like any source: their value comes from v_x or i_x.

7. Ohm's law and resistance [book 2-1]

Formula	In words
$V = IR$	(voltage) = (current)(resistance)
$I = \dfrac{V}{R}$, $\quad R = \dfrac{V}{I}$	rearranged
$I = \dfrac{V_a - V_b}{R}$	current from a to b through R
$G = \dfrac{1}{R}$, $\quad I = GV$	(conductance) = $1 \div$ (resistance)

Ohm's law follows the passive sign convention. If I enters the $-$ side, $I = -V/R$.

Resistance of a shape:

Formula	In words
$R = \dfrac{\rho\ell}{A} = \dfrac{\ell}{\sigma A}$	(resistance) = (resistivity)(length) ÷ (area)
$\ell = \dfrac{RA}{\rho} = R\sigma A$	(length) = (resistance)(area) ÷ (resistivity)
$A = \dfrac{\rho\ell}{R}$	(area) = (resistivity)(length) ÷ (resistance)
$\rho = \dfrac{RA}{\ell}, \quad \sigma = \dfrac{\ell}{RA}$	identify the material from the table

Temperature [book P2.5]:

Formula	In words
$R = R_0(1+\alpha T)$	R_0 at 0°C, T in °C, α in °C^{-1}
$T = \dfrac{R/R_0 - 1}{\alpha}$	(temperature) for a given R
$\dfrac{R_2}{R_1} = \dfrac{1+\alpha T_2}{1+\alpha T_1}$	compare two temperatures

Copper: $\alpha = 4\times10^{-3}$ °C^{-1}. A **thermistor** is built to change a lot with T: $\Delta R = k\Delta T$.

Power and power rating:

Formula	In words
$P = VI = I^2R = \dfrac{V^2}{R} = GV^2$	power in a resistor (always received, becomes heat)
$R = \dfrac{V^2}{P}, \quad R = \dfrac{P}{I^2}$	(resistance) from power
$I_{max} = \sqrt{\dfrac{P_{rated}}{R}}$, $V_{max} = \sqrt{P_{rated}R}$	largest safe current, voltage
fraction $= \dfrac{P_{load}}{P_{source}} = \dfrac{I^2R_L}{VI}$	share of the power that reaches the load

Variable resistors [book Table 2-3]: a **rheostat** (2 terminals) sets 0 to R_{max}. A **potentiometer** (3 terminals) splits R_{max}: $R_{13} + R_{23} = R_{max}$.

Quick example	Work → answer
10 V across 2 kΩ	$I = 5$ mA
100 W lamp at 120 V	$R = 120^2/100 = 144\,\Omega$
2 W in 50 Ω	$I = \sqrt{2/50} = 0.2$ A
$\frac{1}{4}$ W rating, 1 kΩ	$I_{max} = \sqrt{0.25/1000} = 15.8$ mA
carbon film: $\ell = 2$ cm, 1 mm wide, 0.1 mm thick	$A = 10^{-3}\cdot10^{-4} = 10^{-7}$ m^2. $R = \frac{1.40\times10^{-5}\cdot0.02}{10^{-7}} = 2.8\,\Omega$
twin cable 10 m long, AWG-14 copper	both wires: $\ell = 20$ m. $R = \frac{1.72\times10^{-8}\cdot20}{2.01\times10^{-6}} = 0.171\,\Omega$
12 V, cable 0.1 Ω, load 2.9 Ω	$I = 4$ A. $\frac{16\cdot2.9}{12+4} = \frac{46.4}{48} = 0.967$
copper, $R_0 = 100\,\Omega$: R at 50°C	$100(1+0.004\cdot50) = 120\,\Omega$
when is copper R 10% above R_0?	$T = \frac{1.1-1}{0.004} = 25$°C
10 kΩ pot, wiper 30% from terminal 1	$R_{13} = 3$ kΩ, $R_{23} = 7$ kΩ

Wire design chain	Copper, AWG-18, target $R = 0.5\,\Omega$
1. R (from V, I if given)	$R = \dfrac{V}{I}$. Here $0.5\,\Omega$ is given
2. d from the AWG table	1.0 mm $= 1.0\times10^{-3}$ m
3. area	$A = \dfrac{\pi(1.0\times10^{-3})^2}{4} = 7.854\times10^{-7}$ m^2
4. length	$\ell = \dfrac{RA}{\rho} = \dfrac{0.5\cdot7.854\times10^{-7}}{1.72\times10^{-8}} = 22.8$ m

A two-wire cable is twice as long as the run. Current goes out on one wire and back on the other.

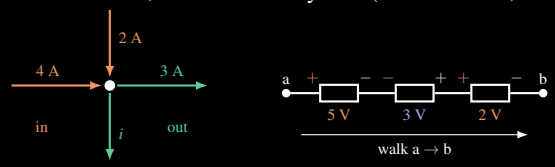
8. Kirchhoff's laws [book 2-2]

Formula	In words
$\sum i_{in} = \sum i_{out}$	KCL: (current into a node) = (current out)
$\sum_n i_n = 0$	KCL, entering = +, leaving = − (or the reverse)
$\sum_n v_n = 0$	KVL: voltages around a closed loop add to 0
$\sum v_{rises} = \sum v_{drops}$	KVL, other form
V_{ab} = sum along any path a→b	path rule: $+v$ if you meet + first

KVL sign rule: go clockwise. Enter an element at +: write $+v$. Enter at −: write $-v$. KCL also holds for a closed boundary around several nodes.

How many equations? [book 2-2.3]

1. Count the unknowns: N.
2. Write M KVL equations. Skip loops through a current source. Each new loop must use an element not used before.
3. Write $N - M$ KCL equations at extraordinary nodes.
4. Put Ohm's law in, then solve the system (Cramer's rule, reference page).



Worked: KCL at the dot (left)

1. PIECES
in: 4 A, 2 A out: 3 A, i

↓

2. TEMPLATE
$\boxed{\text{in}} + \boxed{\text{in}} = \boxed{\text{out}} + \boxed{\text{out}}$

↓

3. PLACE, SOLVE
$4 + 2 = 3 + i \Rightarrow i = 3$ A

↓

4. CHECK
in 6 A, out $3+3 = 6$ A ✓

Worked: V_{ab} by walking the path (right)

1. PIECES (SIGN MET FIRST)
+ first: 5 V, 2 V − first: 3 V

↓

2. TEMPLATE
$V_{ab} = \boxed{+v}\ \boxed{-v} \cdots$ one term per element, in walking order

↓

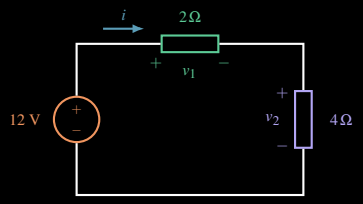
3. PLACE, SOLVE
$V_{ab} = +5 - 3 + 2 = 4$ V

↓

4. CHECK (WALK BACK b → a)
$-2 + 3 - 5 = -4 = V_{ba} = -V_{ab}$ ✓

Quick example	Work → answer
1 A in / 3 A and i out	$i = -2$: 2 A in, against the arrow
10 A in / i_x and $1.5\,i_x$ out	$10 = 2.5\,i_x \Rightarrow i_x = 4$ A
12 V source, 5 V on R_1, find v_2	$-12 + 5 + v_2 = 0 \Rightarrow 7$ V
36 V, 2 Ω, 4 Ω, CCVS $3i$ in one loop	$36 = 2i + 4i + 3i \Rightarrow i = 4$ A
3 unknown currents, 1 current source	M = loops that avoid the source, the rest KCL

A negative answer is fine. The current flows opposite to your arrow.



Worked: KVL clockwise around the loop above

1. PIECES (SIGN MET FIRST, FROM BOTTOM LEFT)
source at $-$ v_1 at $+$ v_2 at $+$

↓

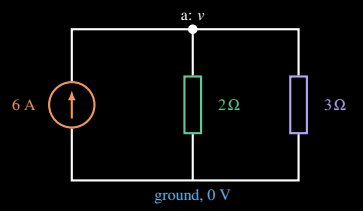
2. TEMPLATE
$\boxed{\pm \text{ source}} + \boxed{\pm v_1} + \boxed{\pm v_2} = 0$, Ohm: $v = iR$

↓

3. PLACE, SOLVE
$-12 + 2i + 4i = 0 \Rightarrow i = 2$ A. $v_1 = 4$ V, $v_2 = 8$ V

↓

4. CHECK (POWER)
$8 + 16 - 24 = 0$ ✓ *(source supplies 24 W)*



Worked: KCL with a node voltage (above)

1. PIECES (STAND AT a, CROSS EACH R)
6 A in $2\,\Omega \to$ ground $3\,\Omega \to$ ground

↓

2. TEMPLATE
$\boxed{\text{in}} = \dfrac{v - \boxed{\text{far}}}{\boxed{R}} + \dfrac{v - \boxed{\text{far}}}{\boxed{R}}$

↓

3. PLACE, SOLVE
$6 = \dfrac{v-0}{2} + \dfrac{v-0}{3} = \dfrac{5v}{6} \Rightarrow v = 7.2$ V

↓

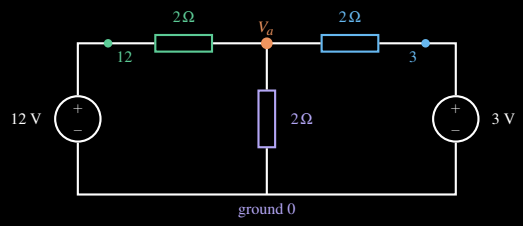
4. CHECK
$\frac{7.2}{2} + \frac{7.2}{3} = 3.6 + 2.4 = 6$ A ✓

8b. nodal analysis: the walk

Unknowns are node voltages. Write one KCL per unknown node. KVL is automatic, since every drop is a difference of node voltages.

Formula	In words
$i = \dfrac{V_{\text{here}} - V_{\text{far}}}{R}$	current *leaving* through one R: stand here, cross it, note where you land
$\sum i_{\text{leaving}} = 0$	one fraction per R touching the node
source $+$ series R	far node $=$ source top $= +v_s$, $-$ on top: $-v_s$
$p_{src} = v_s \lvert i_{\text{series}} \rvert$	leaves its $+$: supplies, enters $+$: absorbs

Steps: (1) ground the node most sources touch. (2) Mark known nodes (source tops). (3) KCL at each unknown, here-node first. (4) Multiply *every term* by the LCD. (5) Solve, then plug back in: each equation $= 0$.



Worked: one unknown node (above)

1. PIECES (STAND AT V_a)
left $2\,\Omega \to 12$ middle $2\,\Omega \to 0$ right $2\,\Omega \to 3$

↓

2. TEMPLATE
$\dfrac{V_a - \boxed{\text{far}}}{\boxed{R}} + \dfrac{V_a - \boxed{\text{far}}}{\boxed{R}} + \dfrac{V_a - \boxed{\text{far}}}{\boxed{R}} = 0$

↓

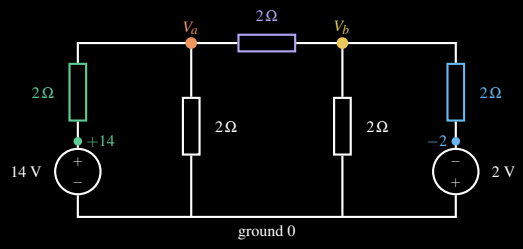
3. PLACE
$\dfrac{V_a - 12}{2} + \dfrac{V_a - 0}{2} + \dfrac{V_a - 3}{2} = 0$

↓

4. $\times 2$ EVERY TERM, SOLVE
$3V_a - 15 = 0 \Rightarrow V_a = 5$ V *(between 3 and 12)*
leaving: -3.5 A *(3.5 A in)*, 2.5 A, 1 A *(sum 0 ✓)*

↓

5. CHECK (POWER): $p_{src} = \boxed{v_s} \cdot \boxed{|i_{\text{series}}|}$, $p_R = i^2 R$
12 V: 3.5 A leaves $+ \Rightarrow 12(3.5) = 42$ W supplied. 3 V: 1 A enters $+ \Rightarrow 3$ W absorbed.
resistors $24.5 + 12.5 + 2 = 39$. $39 + 3 = 42$ ✓



Worked: two unknown nodes (above)

1. PIECES
known: $+14$, -2 *(− on top)*. at V_a: $\to 14$, $\to 0$, $\to V_b$. at V_b: $\to -2$, $\to 0$, $\to V_a$

↓

2. TEMPLATE (ONE KCL PER UNKNOWN NODE)
$\sum \dfrac{V_{\text{here}} - \boxed{\text{far}}}{R} = 0$, one fraction per R touching the node

↓

3. PLACE AT V_a, $\times 2$
$\dfrac{V_a - 14}{2} + \dfrac{V_a - 0}{2} + \dfrac{V_a - V_b}{2} = 0 \Rightarrow 3V_a - V_b = 14$

↓

4. PLACE AT V_b, $\times 2$
$\dfrac{V_b - (-2)}{2} + \dfrac{V_b - 0}{2} + \dfrac{V_b - V_a}{2} = 0 \Rightarrow 3V_b - V_a = -2$

↓

5. SUBSTITUTE THE COEFFICIENT-1 VARIABLE
$V_a = 3V_b + 2 \Rightarrow 3(3V_b + 2) - V_b = 14 \Rightarrow V_b = 1$ V, $V_a = 5$ V

↓

6. CHECK
$3(5) - 1 = 14$ ✓ $3(1) - 5 = -2$ ✓

One node, one voltage: every R touching it goes in the same equation. The shared R appears in both, flipped. Minus before parentheses flips every term: $-(x - 8) = -x + 8$. Labeled $\pm V_x$: $+$ end minus $-$ end.

9. series and parallel resistors [book 2-3]

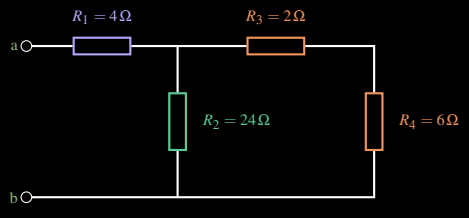
Formula	In words
$R_{eq} = R_1 + R_2 + \cdots$	**series**: resistances add
$\dfrac{1}{R_{eq}} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$	**parallel**: reciprocals add, then flip
$R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$	two in parallel: (product) ÷ (sum)
$\dfrac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3}$	three in parallel
$G_{eq} = G_1 + G_2 + \cdots$	parallel conductances add

Series total > the biggest. Parallel total < the smallest. Two circuits are **equivalent** if they have the same i–v at their terminals.

Combination chart

Pattern	Formula	Example (Ω)
two in series	$R_1 + R_2$	$2 + 3 = 5$
three in series	$R_1 + R_2 + R_3$	$1 + 2 + 3 = 6$
two in parallel	$\dfrac{R_1 R_2}{R_1 + R_2}$	$6 \parallel 3 = \dfrac{18}{9} = 2$
two equal, parallel	$\dfrac{R}{2}$	$10 \parallel 10 = 5$
N equal, parallel	$\dfrac{R}{N}$	four 100s: $\dfrac{100}{4} = 25$
three in parallel (2, 3, 6)	$\dfrac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$	$\dfrac{1}{0.5 + 0.333 + 0.167} = 1$
series, then parallel	$(R_1 + R_2) \parallel R_3$	$(2+4) \parallel 3 = 6 \parallel 3 = 2$
parallel, then series	$(R_1 \parallel R_2) + R_3$	$(6 \parallel 3) + 4 = 2 + 4 = 6$
two series pairs, parallel	$(R_1 + R_2) \parallel (R_3 + R_4)$	$(1+5) \parallel (4+2) = 6 \parallel 6 = 3$
wire across (short)	$R \parallel 0 = 0$	$6 \parallel 0 = 0$
open branch in parallel	$R \parallel \infty = R$	$6 \parallel \infty = 6$
open in series	$R + \infty = \infty$	no current flows
missing R, series	$R = R_{eq} - R_1$	$1000 - 330 = 670$
missing R, parallel	$R = \dfrac{R_{eq} R_1}{R_1 - R_{eq}}$	$R \parallel 12 = 4$: $\dfrac{4 \cdot 12}{12 - 4} = 6$

How to reduce: label the nodes. Merge nodes joined by wires or shorts. Find a clear series pair or parallel pair. Replace it. Redraw. Repeat, starting farthest from the terminals.



Worked: reduce the ladder above to R_{ab}

1. PIECES (FARTHEST FROM a, b FIRST)
R_3, R_4 series R_2 parallel to that pair R_1 series with the rest

↓

2. TEMPLATES
series: $\boxed{R} + \boxed{R}$ parallel: $\dfrac{\boxed{R} \cdot \boxed{R}}{\boxed{R} + \boxed{R}}$

↓

3. PLACE (INSIDE OUT)
$2 + 6 = 8$. $\dfrac{24 \cdot 8}{24 + 8} = 6$. $R_{ab} = 4 + 6 = 10\,\Omega$

↓

4. CHECK
parallel $6 < 8$, the smaller ✓ series $10 > 4$ ✓

Worked: reduce $R_{eq} = 2 + [12 \parallel (4 + 6 \parallel 3)]$ Ω, inside out

1. PIECES (INNERMOST FIRST)
$6 \parallel 3$, then $4 + (\cdot)$, then $12 \parallel (\cdot)$, then $2 + (\cdot)$

↓

2. TEMPLATES
series: $\boxed{R} + \boxed{R}$ parallel: $\dfrac{\boxed{R} \cdot \boxed{R}}{\boxed{R} + \boxed{R}}$

↓

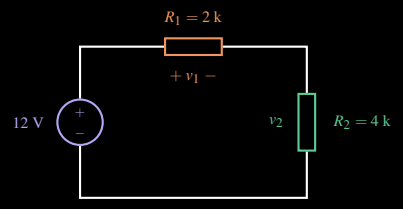
3. PLACE
$\frac{6 \cdot 3}{9} = 2$ → $4 + 2 = 6$ → $\frac{12 \cdot 6}{18} = 4$ → $2 + 4 = 6\,\Omega$

↓

4. CHECK
$12 \parallel 6 = 4$ is below 6, the smaller ✓

10. voltage and current dividers [book 2-3]

Formula	In words
$v_1 = \dfrac{R_1}{R_1 + R_2} v_s$	**voltage divider** (series): own R on top
$v_k = \dfrac{R_k}{R_1 + \cdots + R_N} v_s$	N series resistors
$V_{\text{node}} = \dfrac{R_{\text{below}}}{R_T} v_s$	node voltage (from ground): R_{below} = sum of every R between the node and ground
$v_k = V_{\text{top}} - V_{\text{bottom}}$	across one resistor = difference of its two end nodes
$p_k = (V_{\text{top}} - V_{\text{bottom}})\, I$	series power. All p_k add to $v_s I$
$R_2 = \dfrac{R_1 v_2}{v_s - v_2}$	resistor for a target v_2
$i_1 = \dfrac{R_2}{R_1 + R_2} i_s = \dfrac{G_1}{G_{eq}} i_s$	**current divider** (parallel): other R on top, or own G
$i_k = \dfrac{R_{eq}}{R_k} i_s$	any number of parallel branches
$R_2 = \dfrac{R_1 (i_s - i_2)}{i_2}$	resistor for a target i_2



Worked: voltage divider: v_1 and v_2 above

1. PIECES (FOR v_1)
own: R_1 all series: R_1, R_2 source: 12 V

↓

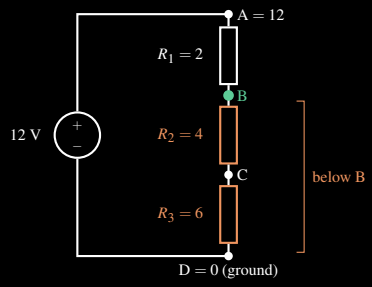
2. TEMPLATE
$v = \dfrac{\boxed{\text{own } R}}{\boxed{\text{every series } R}} \cdot \boxed{\text{source}}$

↓

3. PLACE
$v_1 = \dfrac{2}{2 + 4} \cdot 12 = 4$ V. Own → R_2: $v_2 = \dfrac{4}{6} \cdot 12 = 8$ V

↓

4. CHECK (KVL)
$4 + 8 = 12$, the source ✓



Worked: node voltages in a series stack (above), R in Ω

1. PIECES
$R_T = 12$ between B and ground: R_2, R_3

↓

2. TEMPLATES
$V_{\text{node}} = \dfrac{\boxed{\text{sum of } R \text{ below the node}}}{\boxed{R_T}} \cdot v_s$ $\quad v_k = \boxed{V_{\text{top}}} - \boxed{V_{\text{bottom}}}$

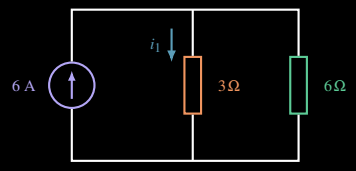
↓

3. PLACE
$V_B = \dfrac{4+6}{12} \cdot 12 = 10$ V, $\quad V_C = \dfrac{6}{12} \cdot 12 = 6$ V, $\quad v_2 = V_B - V_C = 4$ V $(= \frac{4}{12} \cdot 12$ ✓$)$

↓

4. CHECK (POWER): $p = (V_{\text{top}} - V_{\text{bottom}})I$
$I = 1$ A: $\quad p_1 = 2$, $p_2 = 4$, $p_3 = 6$ W. $\quad$ sum $12 = v_s I$ ✓

Node voltage = voltage across only when the other end is ground. $V_B \neq v_2$, but $V_C = v_3$.



Worked: current divider: i_1 above

1. PIECES (FOR i_1)
own: 3Ω $\quad$ other: 6Ω $\quad$ source: 6 A

↓

2. TEMPLATE
$i = \dfrac{\boxed{\text{other } R}}{\boxed{\text{both } R \text{ added}}} \cdot \boxed{i_s}$

↓

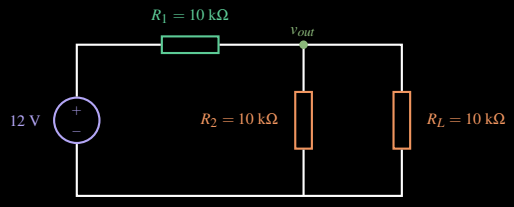
3. PLACE
$i_1 = \dfrac{6}{3+6} \cdot 6 = 4$ A. $\quad i_2 = \dfrac{3}{9} \cdot 6 = 2$ A

↓

4. CHECK
KCL: $4 + 2 = 6$ ✓ $\quad$ same voltage: $4 \cdot 3 = 2 \cdot 6 = 12$ V ✓

Quick example	Work → answer
12 V, $R_1 = 9$ k, want $v_2 = 3$ V	$R_2 = \frac{9 \cdot 3}{12-3} = 3$ kΩ
12 A into 2, 3, 6Ω ($R_{eq} = 1$)	$\frac{1}{2} \cdot 12 = 6$, $\frac{1}{3} \cdot 12 = 4$, $\frac{1}{6} \cdot 12 = 2$ A
10 A, $R_1 = 4$Ω, want $i_2 = 2$ A	$R_2 = \frac{4(10-2)}{2} = 16$Ω

A load across a divider changes it. Combine the load with the resistor it sits across first, then divide.



Worked: the loaded divider above

1. PIECES
R_L across R_2 merge into *one* piece $R_2 \parallel R_L$ $\quad$ top: R_1 $\quad$ source: 12 V

↓

2. TEMPLATE
$v_{out} = \dfrac{\boxed{R_2 \parallel R_L}}{\boxed{R_1} + \boxed{R_2 \parallel R_L}} \cdot \boxed{v_s}$

↓

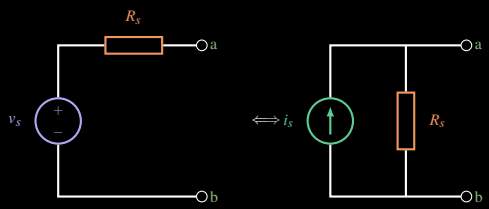
3. PLACE
$10 \parallel 10 = 5$, $\quad v_{out} = \dfrac{5}{10+5} \cdot 12 = 4$ V *(unloaded: 6 V. The load pulls it down)*

↓

4. CHECK (OHM)
$I = \frac{12}{15\text{ k}} = 0.8$ mA, $\quad 0.8$ mA $\cdot 5$ k $= 4$ V ✓

11. source transformation and combining sources [book 2-3]

A voltage source v_s in series with R_s acts exactly like a current source i_s in parallel with the same R_s.



Formula	In words
$i_s = \dfrac{v_s}{R_s}$	voltage source → current source, same R_s in parallel
$v_s = i_s R_s$	current source → voltage source, same R_s in series
arrow of i_s	points toward the + end of v_s
$v_{eq} = v_1 - v_2 + v_3$	voltage sources in series: add with signs
$i_{eq} = i_1 - i_2 + i_3$	current sources in parallel: add with directions

Worked: transform 12 V in series with 4Ω (left → right above)

1. PIECES
$v_s = 12$ V (+ on top) $\quad$ $R_s = 4$Ω in series

↓

2. TEMPLATE
$\boxed{i_s} = \dfrac{\boxed{v_s}}{\boxed{R_s}}$. $\quad$ Same $\boxed{R_s}$, now parallel. The arrow points toward +

↓

3. PLACE
$i_s = \dfrac{12}{4} = 3$ A, pointing up, $\parallel 4$Ω

↓

4. REVERSE (RIGHT → LEFT)
2 A $\parallel$ 5Ω: $v_s = 2 \cdot 5 = 10$ V in series with 5Ω

↓

5. CHECK (SAME TERMINALS)
open a–b: left 12 V, right $3 \cdot 4 = 12$ V ✓ $\quad$ short a–b: left $\frac{12}{4} = 3$ A, right 3 A ✓

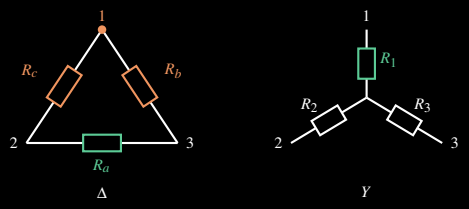
Quick example	Work → answer
9 V and 3 V in series, opposing	6 V
2 A up, 5 A up, 1 A down, in parallel	$2 + 5 - 1 = 6$ A up
open-circuit voltage of a real source	no current, no drop on R_s: $V_{oc} = v_s$

Do not transform the element whose v or i you need. It disappears into the new source.

Unrealizable: different voltage sources in parallel, or different current sources in series. Each node pair has one voltage, each branch one current. This includes dependent sources.

12. wye–delta (Y–Δ, T–Π) [book 2-4]

For networks that are neither series nor parallel (a bridge). Both shapes connect the same three nodes 1, 2, 3. The book's labels: Y arms R_1, R_2, R_3 go to nodes 1, 2, 3. Δ side R_a is opposite node 1 (between 2 and 3), R_b opposite node 2, R_c opposite node 3.



Formula	In words
$R_1 = \dfrac{R_b R_c}{R_a + R_b + R_c}$	$\Delta \to Y$: (Y arm) = (the two Δ sides touching its node, multiplied) ÷ (sum of all three)
$R_2 = \dfrac{R_a R_c}{\Sigma}, \quad R_3 = \dfrac{R_a R_b}{\Sigma}$	same pattern
$R_a = \dfrac{R_1 R_2 + R_2 R_3 + R_1 R_3}{R_1}$	$Y \to \Delta$: (sum of pair products) ÷ (the Y arm at the node opposite)
$R_b = \dfrac{N}{R_2}, \quad R_c = \dfrac{N}{R_3}$	same pattern, N = the numerator above
$R_Y = \dfrac{R_\Delta}{3}, \quad R_\Delta = 3R_Y$	balanced (all equal)

Worked:* $\Delta \to Y$ *with* $R_a = 10$, $R_b = 20$, $R_c = 30\,\Omega$ *(left figure)

1. PIECES (ARM R_1 FIRST)
sides touching node 1: $R_b = 20$, $R_c = 30$ all three: $\Sigma = 60$

↓

2. TEMPLATE
$$R_{\text{arm}} = \frac{\boxed{\text{side touching}} \cdot \boxed{\text{side touching}}}{\boxed{\text{all three added}}}$$

↓

3. PLACE
$$R_1 = \frac{20 \cdot 30}{60} = 10$$
node 2 touches R_a, R_c: $R_2 = \dfrac{10 \cdot 30}{60} = 5$
node 3 touches R_a, R_b: $R_3 = \dfrac{10 \cdot 20}{60} = 3.33\,\Omega$

↓

4. CHECK (NODES 1–2, NODE 3 OPEN)
Y: $R_1 + R_2 = 15$ Δ: $R_c \parallel (R_a + R_b) = 30 \parallel 30 = 15$ ✓

Worked:* $Y \to \Delta$, *reversing the example (right figure)

1. PIECES (SIDE R_a FIRST)
arm at the opposite node (1): $R_1 = 10$ $N = 50 + 16.7 + 33.3 = 100$

↓

2. TEMPLATE
$$R_{\text{side}} = \frac{\boxed{N}}{\boxed{\text{arm at the opposite node}}}$$

↓

3. PLACE
$$R_a = \frac{100}{10} = 10, \quad R_b = \frac{100}{5} = 20, \quad R_c = \frac{100}{3.33} = 30\,\Omega$$

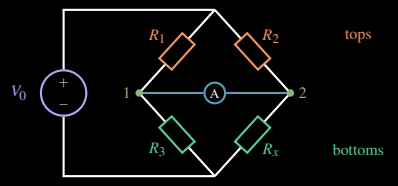
↓

4. CHECK
10, 20, 30: the Δ we started from ✓

Quick example	Work → answer
balanced Δ, each $30\,\Omega$	$R_Y = 10\,\Omega$

Other books swap the labels. Always match by *position*: a Y arm uses the two Δ sides that touch its node.

13. Wheatstone bridge and resistive sensors [book 2-5, 2-6.3]



Formula	In words
$V_1 = \dfrac{R_3}{R_1 + R_3} V_0,$ $V_2 = \dfrac{R_x}{R_2 + R_x} V_0$	each side is a voltage divider
$R_x = \dfrac{R_2}{R_1} R_3$	**balanced** ($I_a = 0$, $V_1 = V_2$): (unknown) from the other three
$V_{out} = \dfrac{V_0 \Delta R}{4R + 2\Delta R}$	sensor bridge: three arms R, one $R + \Delta R$ (exact)
$V_{out} \approx \dfrac{V_0}{4} \cdot \dfrac{\Delta R}{R}$	when $\Delta R \ll R$
$\dfrac{\Delta R}{R} = \dfrac{4 V_{out}}{V_0}$	(relative change) from the reading

***Worked: balance:* $R_1 = 100$, $R_2 = 200$, $R_3 = 150\,\Omega$, *find* R_x**

1. PIECES
tops: 100 (left), 200 (right) bottoms: 150 (left), R_x (right)

↓

2. TEMPLATE (BALANCED: SAME RATIO EACH SIDE)
$$\frac{\boxed{\text{left top}}}{\boxed{\text{left bottom}}} = \frac{\boxed{\text{right top}}}{\boxed{\text{right bottom}}}$$

↓

3. PLACE
$$\frac{100}{150} = \frac{200}{R_x} \Rightarrow R_x = \frac{200 \cdot 150}{100} = 300\,\Omega$$

↓

4. CHECK ($V_1 = V_2$)
$V_1 = \frac{150}{250} V_0 = 0.6 V_0$ $V_2 = \frac{300}{500} V_0 = 0.6 V_0$ ✓

***Worked: sensor bridge:* $V_0 = 5$ V, $\Delta R/R = 0.01$**

1. PIECES
source 5 V the changed arm: $\Delta R/R = 0.01$

↓

2. TEMPLATE
$$V_{out} \approx \frac{\boxed{V_0}}{4} \cdot \boxed{\Delta R/R}$$

↓

3. PLACE
$$V_{out} \approx \frac{5}{4} \cdot 0.01 = 12.5\ \text{mV}$$

↓

4. CHECK (EXACT FORMULA)
$$\frac{V_0 \Delta R}{4R + 2\Delta R} = \frac{5(0.01)}{4.02} = 12.4\ \text{mV}, \text{ close } ✓_{(\Delta R \ll R)}$$

Piezoresistor (pressure sensor):

Formula	In words
$R = R_0(1 + \alpha P), \quad \Delta R = R_0 \alpha P$	P in N/m^2. $P > 0$ stretch, $P < 0$ squeeze
$P = \dfrac{F}{A}$	(pressure) = (force) ÷ (area)
$P = \dfrac{F L}{W H^2}$	cantilever: force F at the tip, length L, width W, thickness H
$V_{out} = \dfrac{V_0}{4} \alpha P = \dfrac{V_0}{4} \alpha \dfrac{F L}{W H^2}$	bridge output for a piezoresistor

Silicon: $\alpha \approx -1 \times 10^{-9}$ m^2/N (usually negative).

Quick example	Work → answer
$V_0 = 4$ V reads 2 mV	$\frac{\Delta R}{R} = \frac{4 \cdot 0.002}{4} = 0.002$
cantilever $F = 10^{-3}$ N, $L = 1$ mm, $W = 0.1$ mm, $H = 10\,\mu$m	$P = \frac{10^{-3} \cdot 10^{-3}}{10^{-4} \cdot 10^{-10}} = 10^{8}$ N/m^2
same, $V_0 = 1$ V, $\alpha = -10^{-9}$	$V_{out} = \frac{1}{4}(-10^{-9})(10^{8}) = -25$ mV

14. nonlinear elements: fuse, diode, LED [book 2-1.4, 2-6]

- **Fuse**: a wire that opens when the current passes its rating. A short at the load draws $I = V_s/R_s$, which blows it.
- **Ideal diode**: a one-way valve. Forward ($V_D > 0$): a short. Reverse ($V_D < 0$): an open, $I = 0$.
- **Real diode model**: off until V_D reaches V_F, then a drop of V_F (plus a small R_D, often ignored). Silicon: $V_F \approx 0.7$ V.
- **LED**: a diode that lights. On fully at about 20 mA. Red $V_F \approx 1.6$ V, green or blue ≈ 3.3 V. Curve: $I \approx aV_D^2$ ($V_D > 0$).

Formula	In words
$I = \dfrac{V_s - V_F}{R}$	forward diode in series with R
$R = \dfrac{V_s - V_F}{I}$	resistor to set an LED current
$R_D \approx \dfrac{\Delta V_D}{\Delta I}$	slope of the curve above V_F

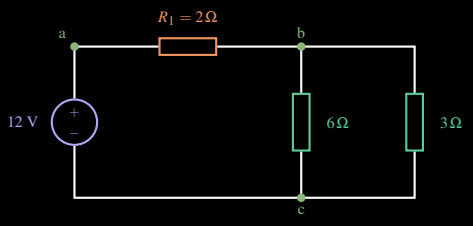
Diode procedure: (1) assume forward bias, (2) compute I, (3) if $I > 0$, done. If not, the diode is off and $I = 0$. For a square or triangle source, do each half of the cycle separately.

Quick example	Work → answer
5 V, 1 kΩ, Si diode forward	$I = \frac{5-0.7}{1000} = 4.3$ mA > 0 ✓
same source reversed (-5 V)	$\frac{-5-0.7}{1000} < 0 \Rightarrow$ off, $I = 0$
red LED at 20 mA from 5 V	$R = \frac{5-1.6}{0.02} = 170\,\Omega$

I–V slopes [book Fig. 2-38]: resistor $1/R$, ideal voltage source vertical (∞), ideal current source flat (0).

15. putting it together: one full circuit

Reduce, find the source current, then expand back out with Ohm's law, KVL, and KCL.



Worked: find every current, voltage, and power

1. PIECES
6, 3 Ω parallel (share b, c) $R_1 = 2\,\Omega$ series 12 V

↓

2. TEMPLATE (REDUCE, THEN EXPAND)
$R_{eq} = \boxed{R_1} + \boxed{6 \parallel 3}$ $i = \dfrac{\boxed{v_s}}{R_{eq}}$ then $v = iR$, KVL, $i = \frac{v}{R}$

↓

3. PLACE (COMBINE, §9)
$6 \parallel 3 = \dfrac{18}{9} = 2\,\Omega$, $R_{eq} = 2 + 2 = 4\,\Omega$, $i = \dfrac{12}{4} = 3$ A

↓

4. DROPS: $v = iR$, THEN KVL
$v_1 = 3 \cdot 2 = 6$ V, $v_{bc} = 12 - 6 = 6$ V *(or $3 \cdot 2$)*

↓

5. BRANCH CURRENTS: $i = v_{bc}/R$
$i_2 = \frac{6}{6} = 1$ A, $i_3 = \frac{6}{3} = 2$ A *(KCL: $1 + 2 = 3$)*

↓

6. CHECK (POWER)
$p_1 = 18$, $p_2 = 6$, $p_3 = 12$ W. Total 36 W $= 12 \cdot 3$ from the source ✓